particles in Tables 1.1 and 1.2. Of course, a string is too small to carry out this experiment literally as described. Rather, by using mathematical descriptions we can theoretically pluck a string. In the mid-1980s, many string adherents believed that the mathematical analysis required for doing this was on the verge of being able to explain every detailed property of the universe on its most microscopic level. Some enthusiastic physicists declared that the T.O.E. had finally been discovered. More than a decade of hindsight has shown that the euphoria generated by this belief was premature. String theory has the makings of a T.O.E., but a number of hurdles remain, preventing us from deducing the spectrum of string vibrations with the precision necessary to compare with experimental results. At the present time, therefore, we do not know if the fundamental characteristics of our universe, summarized in Tables 1.1 and 1.2, can be explained by string theory. As we will discuss in Chapter 9, under certain assumptions that we will clearly state, string theory can give rise to a universe with properties that are in qualitative agreement with the known particle and force data, but extracting detailed numerical predictions from the theory is currently beyond our abilities. And so, although the framework of string theory, unlike that of the point-particle standard model, is capable of giving an explanation for why the particles and forces have the properties they do, we have not, as yet, been able to extract it. But remarkably, string theory is so rich and far-reaching that, even though we cannot yet determine its most detailed properties, we are able to gain insight into a wealth of the new physical phenomena that follow from the theory, as we will see in subsequent chapters.

In the following chapters we shall also discuss the status of the hurdles in some detail, but it is instructive first to understand them at a general level. Strings in the world around us come with a variety of tensions. The string laced through a pair of shoes, for example, is usually quite slack compared to the string stretched from one end of a violin to another. Both of these, in turn, are under far less tension than the steel strings of a piano. The one number that string theory requires in order to set its overall scale is the corresponding tension on its loops. How is this tension determined? Well, if we could pluck a fundamental string we would learn about its stiffness, and in this way we could measure its tension much as is done to measure the tension of